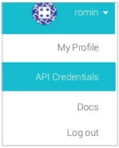

Figure 1: API credentials

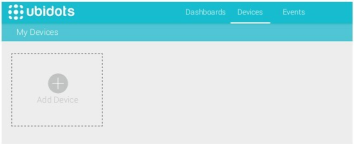

Figure 2: Adding a new device

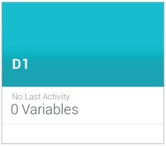

Figure 3: Details about the device

credentials, as shown in Figure 1.

This will lead to a page where you will be shown an API key and a default API token. Carefully note down the API token and store it. From now, we will refer to this as the *API_TOKEN*.

Setting up the device

The first thing that we should do is to set up our devices in Ubidots. The 'Free Plan' comes with support for 20 devices. Think of these devices as those that collect sensor data like temperature. So let's assume that we have deployed a device named D1, from which we are capturing temperature data via a standard temperature sensor. You can have up to 20 devices uniquely configured, which will help you filter, track and measure the data that is coming from various devices, by identifying them via the device's label.

So let's go ahead and set up a device named D1. Follow the steps given below.

1. From the Web portal, visit the *Devices* link at the top. By default, when you go to the *Devices* page, it will show that a default device has been configured. I suggest that you delete it, so that we can start with a clean slate as shown in Figure 2.
2. Click on the *Add Device* icon. Name the device *D1*, as shown in Figure 3.
3. Click on device *D1*. This will lead you to a detailed page, as shown in Figure 4.
4. Now, we need to add a variable. This variable can be the sensor data that we are tracking — for example, temperature. We can capture more than one variable or sensor data coming from a single device. Click on the *Add Variable* icon shown in Figure 4. Select *Default* and then name the variable as *Temperature*, as shown in Figure 5.

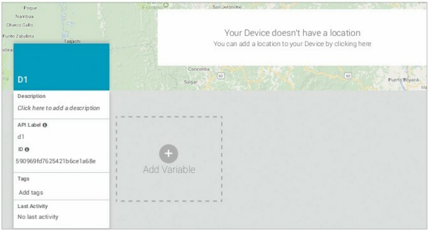

Figure 4: Add a variable for the device

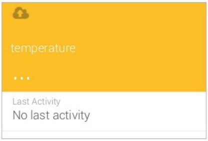

Figure 5: *Temperature* variable

We have successfully created a device *D1* that will provide us temperature values, which we will track via the temperature variable that we created and associate with this device. Let us now look at how we can get our sensor values into the system.

The HTTP API

Ubidots supports both the HTTP and MQTT protocols as a way to ingest data into the system. In this article, we will focus on the HTTP API, and demonstrate how you can ingest data into your application. In our case, we will assume that the sensor data is being generated by device *D1* and the sensor data is a temperature that is being recorded.

The API endpoint is available at *https://things.ubidots.com/api/v1.6/* and we will need to use the *API_TOKEN* as well as our device and variable that we have created to push data into the system.

To send values into the system for a specific device and variable, we need to do a HTTP POST to the HTTPs URL, the format of which is given below: *http://things.ubidots.com/api/v1.6/devices/{LABEL_DEVICE}/{VARIABLE_LABEL}/values*

So for our device *D1*, the {LABEL_DEVICE} will have the value 'D1' and for the {VARIABLE_LABEL}, we shall use the value 'temperature'. In addition, we will need to append the *API_TOKEN* value to the URL as a request parameter.